



US010899752B2

(12) **United States Patent**  
**Sugimoto et al.**(10) **Patent No.:** **US 10,899,752 B2**(45) **Date of Patent:** **\*Jan. 26, 2021**(54) **2,3-DIHYDRO-4H-1,3-BENZOXAZIN-4-ONE  
DERIVATIVES AS MODULATORS OF  
CHOLINERGIC MUSCARINIC M1  
RECEPTOR****C07D 413/10** (2013.01); **C07D 417/06**  
(2013.01); **C07D 417/10** (2013.01); **C07D**  
**471/04** (2013.01); **C07D 491/107** (2013.01);  
**C07F 7/02** (2013.01); **C07F 7/0814** (2013.01);  
**C07B 2200/07** (2013.01)(71) Applicant: **Takeda Pharmaceutical Company  
Limited, Osaka (JP)**(58) **Field of Classification Search**  
CPC ..... **C07D 265/22**; **C07D 413/14**; **A61P 25/28**  
See application file for complete search history.(72) Inventors: **Takahiro Sugimoto, Kanagawa (JP);**  
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**Kenichiro Shimokawa, Kanagawa (JP);**  
**Masataka Murakami, Kanagawa (JP);**  
**Jinichi Yonemori, Kanagawa (JP);**  
**Takuto Kojima, Kanagawa (JP)**(56) **References Cited**

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(Continued)

## FOREIGN PATENT DOCUMENTS

JP 2001-131173 5/2001  
WO 94/00448 1/1994  
(Continued)(73) Assignee: **Takeda Pharmaceutical Company  
Limited, Osaka (JP)**(\*) Notice: Subject to any disclaimer, the term of this  
patent is extended or adjusted under 35  
U.S.C. 154(b) by 0 days.This patent is subject to a terminal dis-  
claimer.(21) Appl. No.: **16/394,762**(22) Filed: **Apr. 25, 2019**(65) **Prior Publication Data**

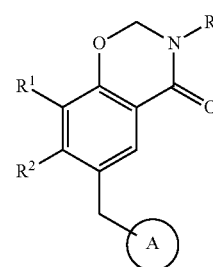
US 2019/0248776 A1 Aug. 15, 2019

**Related U.S. Application Data**(63) Continuation of application No. 15/735,418, filed as  
application No. PCT/JP2016/069189 on Jun. 23,  
2016, now Pat. No. 10,323,027.(30) **Foreign Application Priority Data**Jun. 26, 2015 (JP) ..... 2015-129043  
Oct. 20, 2015 (JP) ..... 2015-206797(51) **Int. Cl.****C07D 413/14** (2006.01)  
**C07D 413/04** (2006.01)  
**C07D 413/06** (2006.01)  
**C07D 413/10** (2006.01)  
**C07D 417/06** (2006.01)  
**C07D 471/04** (2006.01)  
**C07D 491/107** (2006.01)  
**C07D 265/22** (2006.01)  
**C07F 7/02** (2006.01)  
**A61P 25/28** (2006.01)  
**C07D 417/10** (2006.01)  
**C07F 7/08** (2006.01)(52) **U.S. Cl.**CPC ..... **C07D 413/14** (2013.01); **A61P 25/28**  
(2018.01); **C07D 265/22** (2013.01); **C07D**  
**413/04** (2013.01); **C07D 413/06** (2013.01);

## OTHER PUBLICATIONS

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Oct. 6, 2016, 5 pages.Kuduk, Scott D. et al. Novel M1 allosteric ligands: a patent review,  
*Expert Opinion on Therapeutic Patents*, vol. 22, No. 12, Oct. 23,  
2012, pp. 1385-1398. Online [retrieved on Sep. 22, 2016] retrieved  
at: <http://www.tandfonline.com/loi/ietp20> , 15 pages.

(Continued)

*Primary Examiner* — Rei Tsang Shiao(74) *Attorney, Agent, or Firm* — Hamre, Schumann,  
Mueller & Larson, P.C.(57) **ABSTRACT**The present invention provides a compound having a cho-  
linergic muscarinic M1 receptor positive allosteric modula-  
tor activity, which may be useful as a prophylactic or  
therapeutic drug for Alzheimer's disease, schizophrenia,  
pain, sleep disorder, Parkinson's disease dementia, dementia  
with Lewy bodies and the like. The present invention relates  
to a compound represented by the formula (I) or a salt  
thereof:

(I)

wherein each symbol is as defined in the attached  
DESCRIPTION.**11 Claims, No Drawings**

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**2,3-DIHYDRO-4H-1,3-BENZOXAZIN-4-ONE  
DERIVATIVES AS MODULATORS OF  
CHOLINERGIC MUSCARINIC M1  
RECEPTOR**

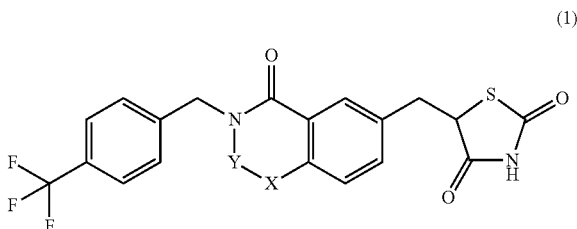
**TECHNICAL FIELD**

The present invention relates to a heterocyclic compound which has a cholinergic muscarinic M1 receptor positive allosteric modulator activity and may be useful as a medicament such as an agent for the prophylaxis or treatment of Alzheimer's disease, schizophrenia, pain, sleep disorder, Parkinson's disease dementia, dementia with Lewy bodies and the like. As used herein, the positive allosteric modulator activity refers to an action to potentiate receptor function by binding at a different site from that of an endogenous activator (acetylcholine for this receptor).

**BACKGROUND OF THE INVENTION**

Acetylcholine is a neurotransmitter that induces signal transduction in the central nervous system and the neuromuscular connections (the parasympathetic nerve and motor nerve). In the central nervous system, nuclei of origin of the acetylcholine neuron are in the brain stem and forebrain, and those acetylcholine neurons project to cerebral cortex, hippocampus, and limbic area. In addition, some interneurons in some brain areas such as striatum utilize acetylcholine as a neurotransmitter. Acetylcholine receptor is classified into a ligand dependent-ion channel (cholinergic nicotinic receptor) and a G-protein-coupled receptor (cholinergic muscarinic receptor). The cholinergic muscarinic receptor is one kind of receptor for excitatory neurotransmitter, acetylcholine, and was named based on the selective activation of the receptor by muscarine. The muscarinic receptor is further classified into subtypes of M1 to M5. The M1 receptor is known to be mainly distributed in the brain, and deeply involved particularly in learning, memory, sleep, neuropathic pain, and the like. The importance of cholinergic muscarinic M1 receptor in brain physiology is well known, and a compound which enhances M1 receptor function is expected to be useful as an agent for the prophylaxis or treatment of mental diseases, neurodegenerative diseases, memory disorders, pain, sleep disorders, Parkinson's disease dementia, dementia with Lewy bodies and the like (non-patent document 1).

Patent document 1 discloses the following compound that binds as a ligand of human peroxisome proliferator-activated receptor  $\gamma$  (PPAR $\gamma$ ) to the receptor and activates same to show a strong hypoglycemic action.

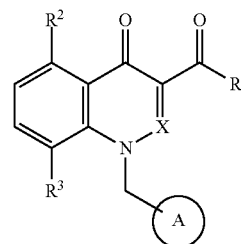


wherein each symbol is as defined in the document.

Patent document 2 discloses the following compound, which has a cholinergic muscarinic M1 receptor positive allosteric modulator activity and is useful as a prophylactic

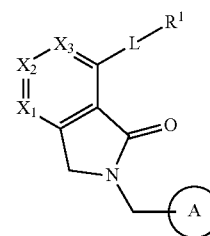
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or therapeutic drug for Alzheimer's disease, schizophrenia, pain, sleep disorder and the like.



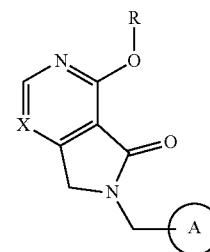
wherein each symbol is as defined in the document.

Patent document 3 discloses the following compound, which has a cholinergic muscarinic M1 receptor positive allosteric modulator activity and is useful as a prophylactic or therapeutic drug for Alzheimer's disease, schizophrenia, pain, sleep disorder and the like.



wherein each symbol is as defined in the document.

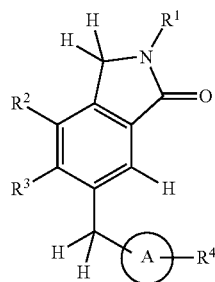
Patent document 4 discloses the following compound, which has a cholinergic muscarinic M1 receptor positive allosteric modulator activity and is useful as a prophylactic or therapeutic drug for Alzheimer's disease, schizophrenia, pain, sleep disorder, Parkinson's disease dementia, dementia with Lewy bodies and the like.



wherein each symbol is as defined in the document.

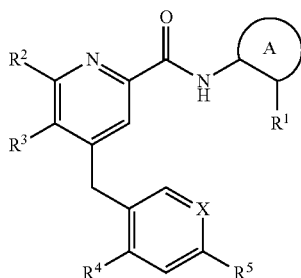
Patent document 5 discloses the following compound, which has a cholinergic muscarinic M1 receptor positive allosteric modulator activity and is useful as a prophylactic or therapeutic drug for Alzheimer's disease, schizophrenia, pain, sleep disorder, Parkinson's disease dementia, dementia with Lewy bodies and the like.

3



wherein each symbol is as defined in the document.

Patent document 6 discloses the following compound, which has a cholinergic muscarinic M1 receptor positive allosteric modulator activity and is useful as a prophylactic or therapeutic drug for Alzheimer's disease, schizophrenia, pain, sleep disorder, Parkinson's disease dementia, dementia with Lewy bodies and the like.



wherein each symbol is as defined in the document.

## DOCUMENT LIST

## Patent Documents

patent document 1: JP-A-2001-131173  
 patent document 2: WO 2013/129622  
 patent document 3: WO 2014/077401  
 patent document 4: WO 2015/174534  
 patent document 5: WO 2015/163485  
 patent document 6: WO 2015/190564

## Non-Patent Document

non-patent document 1: Nature Reviews Drug Discovery, 2007, 6, 721-733.

## SUMMARY OF THE INVENTION

## Problems to be Solved by the Invention

The development of a compound having a cholinergic muscarinic M1 receptor (M1 receptor) positive allosteric modulator activity and useful as an agent for the prophylaxis or treatment of for Alzheimer's disease, schizophrenia, pain, sleep disorder, Parkinson's disease dementia, dementia with Lewy bodies and the like is desired.

## Means of Solving the Problems

The present inventors have conducted intensive studies in an attempt to solve the aforementioned problems and found

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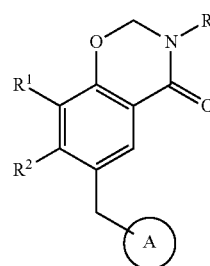
(I) that a compound represented by the following formula (I) may have a cholinergic muscarinic M1 receptor positive allosteric modulator activity, which resulted in the completion of the present invention.

5 Accordingly, the present invention relates to the following.

[1] A compound represented by the formula

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(I)



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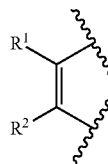
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wherein

(I) R<sup>1</sup> and R<sup>2</sup> are each independently a hydrogen atom or a substituent, or a partial structure:

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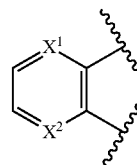
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is optionally

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45

(wherein X<sup>1</sup> and X<sup>2</sup> are each independently CH or N),

R<sup>3</sup> is a hydrogen atom, an optionally substituted C<sub>1-6</sub> alkyl group or an optionally substituted cyclic group, and

ring A is a ring which is optionally further substituted, excluding 6-[(2,4-dioxothiazolidin-5-yl)methyl]-3-[[[(4-trifluoromethyl)phenyl)methyl]-1,3-benzoxazin-4-one, or a salt thereof (sometimes to be abbreviated as "compound (I)" in the present specification).

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[2] The compound described in [1], wherein R<sup>3</sup> is (1) a hydrogen atom, (2) a C<sub>1-6</sub> alkyl group optionally substituted by 1 to 3 substituents selected from a hydroxy group and a 3- to 14-membered non-aromatic heterocyclic group, (3) a C<sub>3-10</sub> cycloalkyl group optionally substituted by 1 to 3 hydroxy groups, (4) a C<sub>6-14</sub> aryl group optionally substituted by 1 to 3 halogen atoms or (5) a 3- to 14-membered non-aromatic heterocyclic group optionally substituted by 1 to 3 hydroxy groups, or a salt thereof.

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[3] The compound described in [1], wherein the ring A is an optionally further substituted 6-membered aromatic ring, or a salt thereof.

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heat, light, a radical initiator such as benzoyl peroxide, azobisisobutyronitrile and the like to the reaction system.

When halogenation reaction of a hydroxy group is carried out in each step, examples of the halogenating agent to be used include hydrohalic acids and acid halides of inorganic acids, specifically, hydrochloric acid, thionyl chloride, phosphorus oxychloride and the like for chlorination, 48% hydrobromic acid and the like for bromination. In addition, a method of producing an alkyl halide by reacting an alcohol with triphenylphosphine and carbon tetrachloride or carbon tetrabromide or the like can be employed. Alternatively, a method of producing an alkyl halide via two step comprising converting an alcohol to the corresponding sulfonate ester, and then reacting the sulfonate ester with lithium bromide, lithium chloride or sodium iodide can also be employed.

When Arbuzov reaction is carried out in each step, examples of the reagent to be used include alkyl halides such as ethyl bromoacetate and the like; and phosphites such as triethyl phosphite, tri(isopropyl) phosphite and the like.

When sulfonate esterification reaction is carried out in each step, examples of the sulfonating agent to be used include methanesulfonyl chloride, p-toluenesulfonyl chloride, methanesulfonic anhydride, p-toluenesulfonic anhydride, N-phenylbis(trifluoromethanesulfonylimide) and the like.

When hydrolysis reaction is carried out in each step, an acid or a base is used as a reagent. For acid hydrolysis reaction of tert-butyl ester, formic acid, triethylsilane and the like may be added to reductively-trap tert-butyl cation which is by-produced.

When dehydration reaction is carried out in each step, examples of the dehydrating agent to be used include sulfuric acid, diphosphorus pentaoxide, phosphorus oxychloride, N,N'-dicyclohexylcarbodiimide, alumina, polyphosphoric acid and the like.

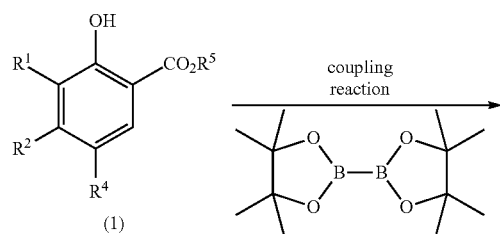
When oxazinone ring formation reaction of salicylamide is carried out in each step, examples of the reagent to be used include formaldehyde, paraformaldehyde, 1,3,5-trioxane, dibromomethane, diiodomethane, chloriodomethane and the like. Examples of the acid to be used include formic acid, p-toluenesulfonic acid, camphorsulfonic acid and the like, and examples of the base to be used include cesium carbonate and the like.

When ester-amide exchange reaction is carried out in each step, examples of the base to be used include triethylamine, N,N-diisopropylethylamine, 1,8-diazabicyclo[5.4.0]-7-undecene, sodium hydrogen carbonate and the like.

When halogenation reaction is carried out in each step, examples of the halogenating agent to be used include N-chlorosuccinimide (NCS), sulfuryl chloride and the like for chlorination, and N-bromosuccinimide (NBS), bromine and the like for bromination.

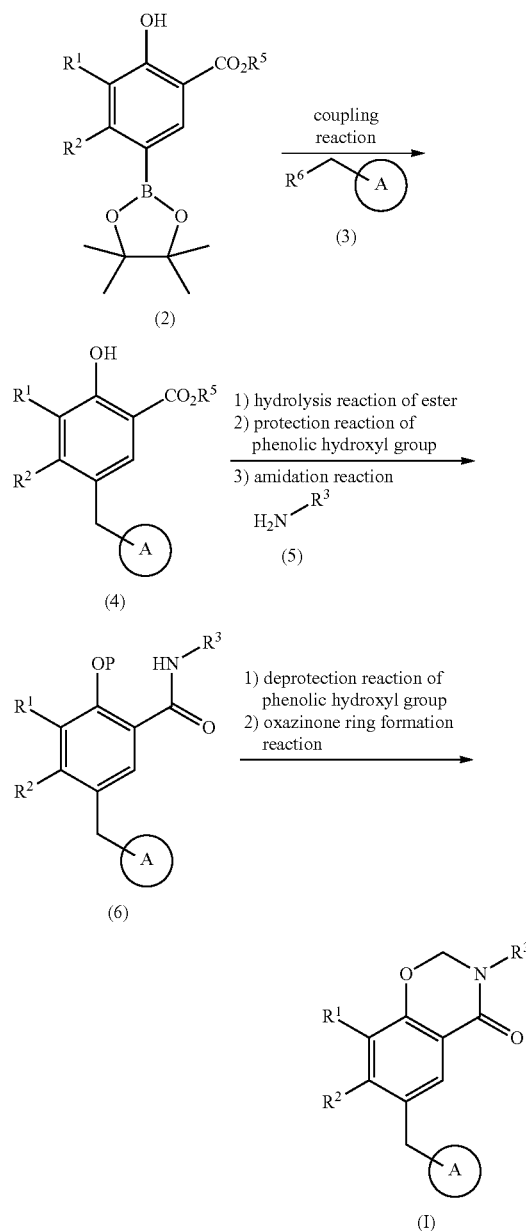
Compound (I) can be produced by the methods shown in the following schemes or a method analogous thereto or a method described in the Examples.

Compound (I) can be produced from compound (1) by the following method.



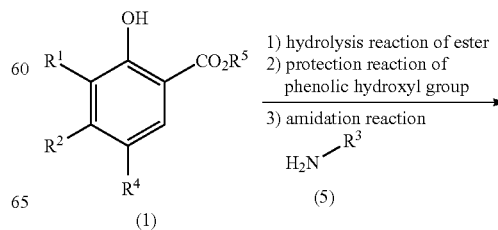
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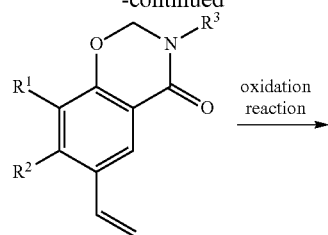
wherein R<sup>4</sup> and R<sup>6</sup> are halogen atoms, R<sup>5</sup> is a lower alkyl group, P is a protecting group such as a benzoyl group and the like, and ring A and R<sup>1</sup>-R<sup>3</sup> are as defined above.

Compound (I) can also be produced from compound (1) by the following method.

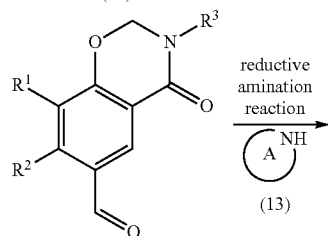


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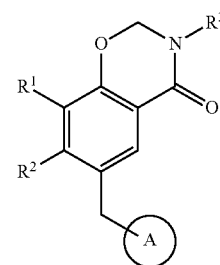


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(13)

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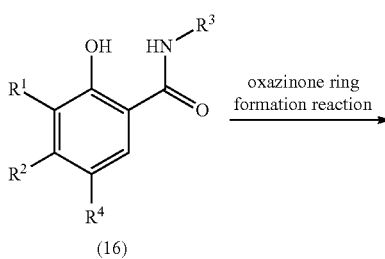
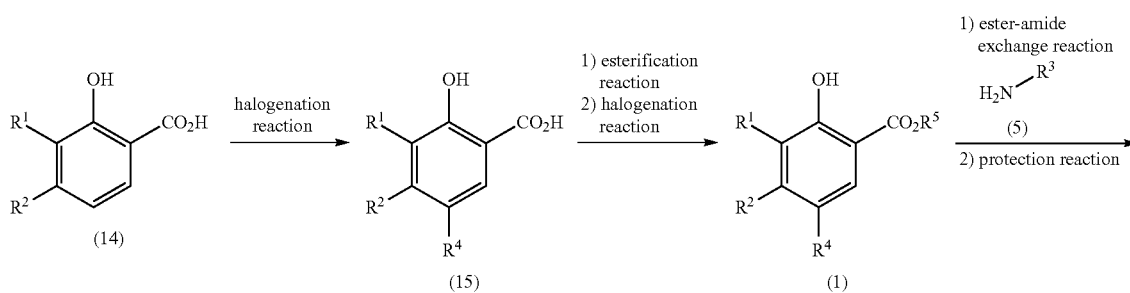


(I)

wherein ring A, R<sup>1</sup>-R<sup>5</sup> and P are as defined above.

In the production of compound (I), a step for protection and/or deprotection of a phenolic hydroxyl group in the above-mentioned scheme may not be necessary.

Compound (I) can also be produced from compound (14) by the following method.



(16)

TABLE 1-continued

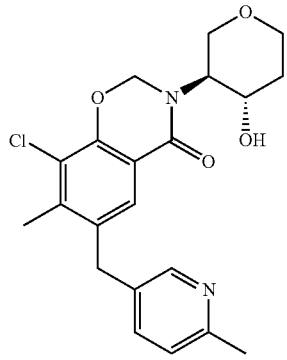
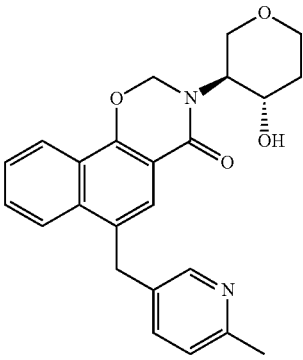
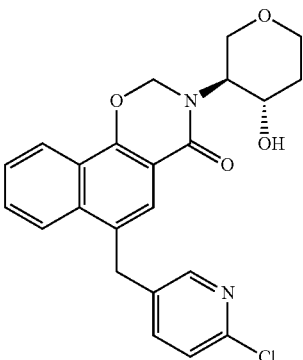
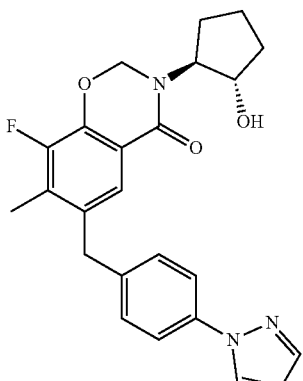
| Ex.<br>No. | IUPAC name                                                                                                                               | structural<br>formula                                                               | salt | MS    |
|------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|-------|
| 2          | 8-chloro-3-((1S,2S)-2-hydroxycyclohexyl)-7-methyl-6-((6-methylpyridin-3-yl)methyl)-2,3-dihydro-4H-1,3-benzoxazin-4-one                   |    |      | 401.1 |
| 3          | 1,5-anhydro-2,4-dideoxy-2-(6-((6-methylpyridin-3-yl)methyl)-4-oxo-2H-naphtho[2,1-e][1,3]oxazin-3(4H)-yl)-threo-pentitol (optical isomer) |   | HCl  | 405.1 |
| 4          | 1,5-anhydro-2-(6-((6-chloropyridin-3-yl)methyl)-4-oxo-2H-naphtho[2,1-e][1,3]oxazin-3(4H)-yl)-2,4-dideoxy-L-threo-pentitol                |  |      | 425.1 |
| 5          | 8-fluoro-3-((1S,2S)-2-hydroxycyclopentyl)-7-methyl-6-(4-(1H-pyrazol-1-yl)benzyl)-2,3-dihydro-4H-1,3-benzoxazin-4-one                     |  |      | 422.2 |

TABLE 1-continued

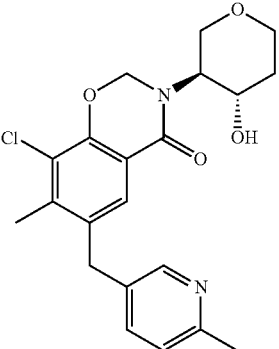
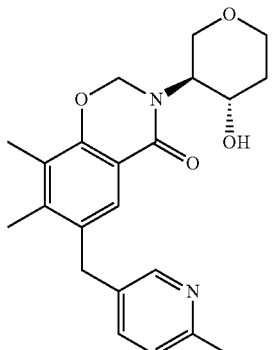
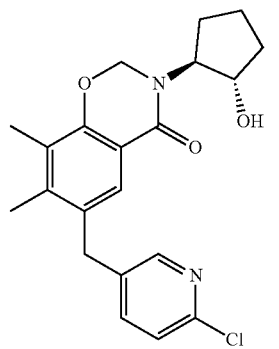
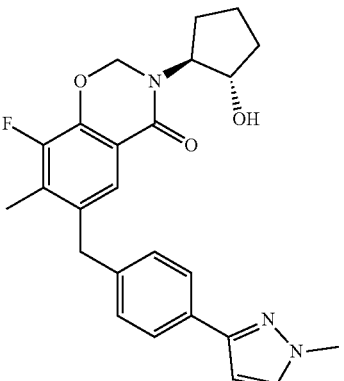
| Ex.<br>No. | IUPAC name                                                                                                                                 | structural<br>formula                                                               | salt | MS    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------|-------|
| 6          | 1,5-anhydro-2-(8-chloro-7-methyl-6-((6-methylpyridin-3-yl)methyl)-4-oxo-2H-1,3-benzoxazin-3(4H)-yl)-2,4-dideoxy-L-threo-pentitol           |    |      | 403.1 |
| 7          | 1,5-anhydro-2,4-dideoxy-2-(7,8-dimethyl-6-((6-methylpyridin-3-yl)methyl)-4-oxo-2H-1,3-benzoxazin-3(4H)-yl)-threo-pentitol (optical isomer) |   | HCl  | 383.2 |
| 8          | 6-((6-chloropyridin-3-yl)methyl)-3-((1S,2S)-2-hydroxycyclopentyl)-7,8-dimethyl-2,3-dihydro-4H-1,3-benzoxazin-4-one                         |  |      | 387.1 |
| 9          | 8-fluoro-3-((1S,2S)-2-hydroxycyclopentyl)-7-methyl-6-(4-(1-methyl-1H-pyrazol-3-yl)benzyl)-2,3-dihydro-4H-1,3-benzoxazin-4-one              |  |      | 436.2 |

TABLE 1-continued

| Ex.<br>No. | IUPAC name                                                                                                                         | structural<br>formula | salt | MS    |
|------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------|------|-------|
| 10         | 1,5-anhydro-2-((6-chloropyridin-3-yl)methyl)-7,8-dimethyl-4-oxo-2H-1,3-benzoxazin-3(4H)-yl)-2,4-dideoxy-L-threo-pentitol           |                       |      | 403.1 |
| 11         | 3-((1S,2S)-2-hydroxycyclopentyl)-7,8-dimethyl-6-((6-methylpyridin-3-yl)methyl)-2,3-dihydro-4H-1,3-benzoxazin-4-one                 |                       |      | 367.2 |
| 12         | 4-((8-fluoro-3-((1S,2S)-2-hydroxycyclopentyl)-7-methyl-4-oxo-3,4-dihydro-2H-1,3-benzoxazin-6-yl)methyl)benzonitrile                |                       |      | 381.1 |
| 13         | 8-fluoro-3-((1S,2S)-2-hydroxycyclopentyl)-7-methyl-6-((6-(1H-pyrazol-1-yl)pyridin-3-yl)methyl)-2,3-dihydro-4H-1,3-benzoxazin-4-one |                       |      | 423.0 |



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Example No. 104 (3 mg/kg)-scopolamine group:  
61.6±2.2%

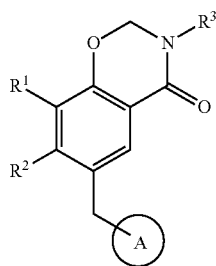
## INDUSTRIAL APPLICABILITY

The compound of the present invention may be useful as a cholinergic muscarinic M1 receptor positive allosteric modulator, or a medicament such as an agent for the prophylaxis or treatment of Alzheimer's disease, schizophrenia, pain, sleep disorder, Parkinson's disease dementia, dementia with Lewy bodies and the like.

This application is based on patent application Nos. 2015-129043 and 2015-206797 filed in Japan, the contents of which are encompassed in full herein.

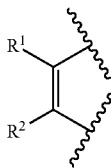
The invention claimed is:

1. A method for the treatment of Alzheimer's disease, schizophrenia, cognitive impairment associate with schizophrenia, pain, sleep disorder, Parkinson's disease dementia, Parkinson's disease, or dementia with Lewy bodies in a mammal, which comprises administering to the mammal an effective amount of a compound represented by the formula

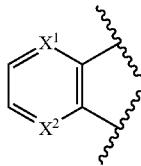


wherein

R<sup>1</sup> and R<sup>2</sup> are each independently a hydrogen atom or a substituent, or  
a partial structure:



is optionally



(wherein X<sup>1</sup> and X<sup>2</sup> are each independently CH or N),

R<sup>3</sup> is a hydrogen atom, an optionally substituted C<sub>1-6</sub> alkyl group or an optionally substituted cyclic group, and  
ring A is a ring which is optionally further substituted, excluding 6-[(2,4-dioxothiazolidin-5-yl)methyl]-3-[(4-trifluoromethyl)phenyl)methyl]-1,3-benzoxazin-4-one,  
or a salt thereof.

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2. The method according to claim 1, wherein R<sup>3</sup> is (1) a hydrogen atom, (2) a C<sub>1-6</sub> alkyl group optionally substituted by 1 to 3 substituents selected from a hydroxy group and a 3- to 14-membered non-aromatic heterocyclic group, (3) a C<sub>3-10</sub> cycloalkyl group optionally substituted by 1 to 3 hydroxy groups, (4) a C<sub>6-14</sub> aryl group optionally substituted by 1 to 3 halogen atoms or (5) a 3- to 14-membered non-aromatic heterocyclic group optionally substituted by 1 to 3 hydroxy groups.

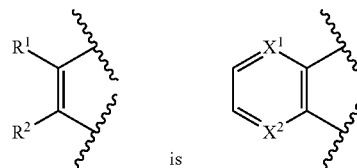
3. The method according to claim 1, wherein the ring A is an optionally further substituted 6-membered aromatic ring.

4. The method according to claim 1, wherein the ring A is a 6-membered aromatic ring optionally further substituted by 1 to 5 substituents selected from

- (i) a halogen atom,
- (ii) a cyano group,
- (iii) a C<sub>1-6</sub> alkyl group,
- (iv) a C<sub>2-6</sub> alkenyl group,
- (v) a C<sub>2-6</sub> alkynyl group optionally substituted by a tri-C<sub>1-6</sub> alkylsilyl group,
- (vi) a C<sub>1-6</sub> alkoxy group and
- (vii) a 5- to 14-membered aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from a halogen atom and a C<sub>1-6</sub> alkyl group.

5. The method according to claim 1, wherein at least one of R<sup>1</sup> and R<sup>2</sup> is a substituent.

6. The method according to claim 1, wherein R<sup>1</sup> is a hydrogen atom, a halogen atom or a C<sub>1-6</sub> alkyl group; R<sup>2</sup> is a hydrogen atom, a halogen atom, a C<sub>1-6</sub> alkyl group, a C<sub>2-6</sub> alkenyl group or a C<sub>1-6</sub> alkoxy group; or the partial structure:

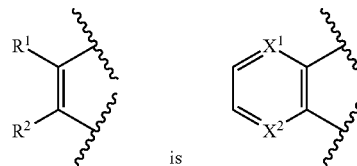


(wherein X<sup>1</sup> and X<sup>2</sup> are each CH);

R<sup>3</sup> is a hydrogen atom, an optionally substituted C<sub>1-6</sub> alkyl group, an optionally substituted C<sub>3-10</sub> cycloalkyl group, an optionally substituted C<sub>6-14</sub> aryl group or an optionally substituted 3- to 14-membered non-aromatic heterocyclic group; and

ring A is a C<sub>6-14</sub> aromatic hydrocarbon ring, a 3- to 14-membered non-aromatic heterocycle or a 5- to 14-membered aromatic heterocycle, each of which is optionally further substituted.

7. The method according to claim 1, wherein R<sup>1</sup> is a hydrogen atom, a halogen atom or a C<sub>1-6</sub> alkyl group; R<sup>2</sup> is a hydrogen atom, a halogen atom, a C<sub>1-6</sub> alkyl group, a C<sub>2-6</sub> alkenyl group or a C<sub>1-6</sub> alkoxy group; or the partial structure:



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(wherein  $X^1$  and  $X^2$  are each CH);

$R^3$  is

- (1) a hydrogen atom,
- (2) a  $C_{1-6}$  alkyl group optionally substituted by 1 to 3 substituents selected from a hydroxy group and a 3- to 14-membered non-aromatic heterocyclic group,
- (3) a  $C_{3-10}$  cycloalkyl group optionally substituted by 1 to 3 hydroxy groups,
- (4) a  $C_{6-14}$  aryl group optionally substituted by 1 to 3 halogen atoms or

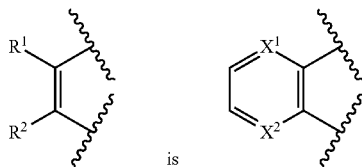
- (5) a 3- to 14-membered non-aromatic heterocyclic group optionally substituted by 1 to 3 hydroxy groups; and

ring A is a  $C_{6-14}$  aromatic hydrocarbon ring, a 3- to 14-membered non-aromatic heterocycle or a 5- to 14-membered aromatic heterocycle, each of which is optionally further substituted by 1 to 5 substituents selected from

- (i) a halogen atom,
- (ii) a cyano group,
- (iii) an oxo group,
- (iv) a  $C_{1-6}$  alkyl group optionally substituted by 1 to 3 halogen atoms,
- (v) a  $C_{2-6}$  alkenyl group,
- (vi) a  $C_{2-6}$  alkynyl group optionally substituted by a tri- $C_{1-6}$  alkylsilyl group,
- (vii) a  $C_{1-6}$  alkoxy group,
- (viii) a  $C_{1-6}$  alkoxy-carbonyl group and
- (ix) a 5- to 14-membered aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from a halogen atom and a  $C_{1-6}$  alkyl group.

8. The method according to claim 1, wherein  $R^1$  is a hydrogen atom, a halogen atom or a  $C_{1-6}$  alkyl group;

$R^2$  is a hydrogen atom, a halogen atom, a  $C_{1-6}$  alkyl group, a  $C_{2-6}$  alkenyl group or a  $C_{1-6}$  alkoxy group; or the partial structure:



(wherein  $X^1$  and  $X^2$  are each CH);

$R^3$  is

- (1) a hydrogen atom,
- (2) a  $C_{1-6}$  alkyl group optionally substituted by 1 to 3 substituents selected from a hydroxy group and a 3- to 14-membered non-aromatic heterocyclic group,
- (3) a  $C_{3-10}$  cycloalkyl group optionally substituted by 1 to 3 hydroxy groups,
- (4) a  $C_{6-14}$  aryl group optionally substituted by 1 to 3 halogen atoms or

- (5) a 3- to 14-membered non-aromatic heterocyclic group optionally substituted by 1 to 3 hydroxy groups; and

ring A is a 6-membered aromatic ring optionally further substituted by 1 to 5 substituents selected from

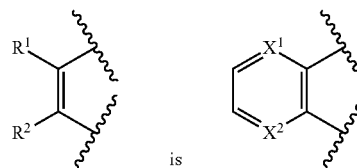
- (i) a halogen atom,
- (ii) a cyano group,
- (iii) a  $C_{1-6}$  alkyl group,
- (iv) a  $C_{2-6}$  alkenyl group,
- (v) a  $C_{2-6}$  alkynyl group optionally substituted by a tri- $C_{1-6}$  alkylsilyl group,
- (vi) a  $C_{1-6}$  alkoxy group and

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(vii) a 5- to 14-membered aromatic heterocyclic group optionally substituted by 1 to 3 substituents selected from a halogen atom and a  $C_{1-6}$  alkyl group.

9. The method according to claim 1, wherein  $R^1$  is a hydrogen atom, a halogen atom or a  $C_{1-6}$  alkyl group;

$R^2$  is a  $C_{1-6}$  alkyl group, a  $C_{2-6}$  alkenyl group or a  $C_{1-6}$  alkoxy group; or the partial structure:



(wherein  $X^1$  and  $X^2$  are each CH);

$R^3$  is

- (1) a  $C_{3-10}$  cycloalkyl group optionally substituted by 1 to 3 hydroxy groups or
- (2) a 3- to 14-membered non-aromatic heterocyclic group optionally substituted by 1 to 3 hydroxy groups; and

ring A is a  $C_{6-14}$  aromatic hydrocarbon ring, a 3- to 14-membered non-aromatic heterocycle or a 5- to 14-membered aromatic heterocycle, each of which is optionally further substituted by 1 to 5 substituents selected from

- (i) a halogen atom,
- (ii) a cyano group,
- (iii) a  $C_{1-6}$  alkyl group,
- (iv) a  $C_{1-6}$  alkoxy group and
- (v) a 5- to 14-membered aromatic heterocyclic group optionally substituted by 1 to 3  $C_{1-6}$  alkyl groups.

10. The method according to claim 1, wherein  $R^1$  is a hydrogen atom, a halogen atom or a  $C_{1-6}$  alkyl group;

$R^2$  is a  $C_{1-6}$  alkyl group, a  $C_{2-6}$  alkenyl group or a  $C_{1-6}$  alkoxy group;

$R^3$  is

- (1) a  $C_{3-10}$  cycloalkyl group optionally substituted by 1 to 3 hydroxy groups or
- (2) a 3- to 14-membered non-aromatic heterocyclic group optionally substituted by 1 to 3 hydroxy groups; and

ring A is a  $C_{6-14}$  aromatic hydrocarbon ring, a 3- to 14-membered non-aromatic heterocycle or a 5- to 14-membered aromatic heterocycle, each of which is optionally further substituted by 1 to 5 substituents selected from

- (i) a halogen atom,
- (ii) a cyano group,
- (iii) a  $C_{1-6}$  alkyl group,
- (iv) a  $C_{1-6}$  alkoxy group and
- (v) a 5- to 14-membered aromatic heterocyclic group optionally substituted by 1 to 3  $C_{1-6}$  alkyl groups.

11. The method according to claim 1, wherein  $R^1$  is a hydrogen atom or a halogen atom;

$R^2$  is a  $C_{1-6}$  alkyl group;

$R^3$  is

- (1) a  $C_{3-10}$  cycloalkyl group optionally substituted by 1 to 3 hydroxy groups or
- (2) a 3- to 14-membered non-aromatic heterocyclic group optionally substituted by 1 to 3 hydroxy groups; and

ring A is a  $C_{6-14}$  aromatic hydrocarbon ring or a 5- to 14-membered aromatic heterocycle, each of which is optionally further substituted by 1 to 5 substituents selected from